TorchJD: Jacobian Descent in PyTorch

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GitHub
github.com/SimplexLab/TorchJD

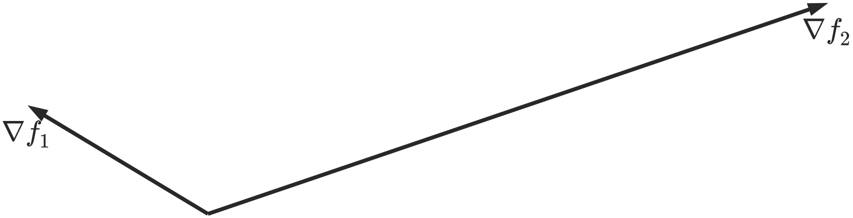
TorchJD
torchjd.org

arXiv
arxiv.org/pdf/2406.16232

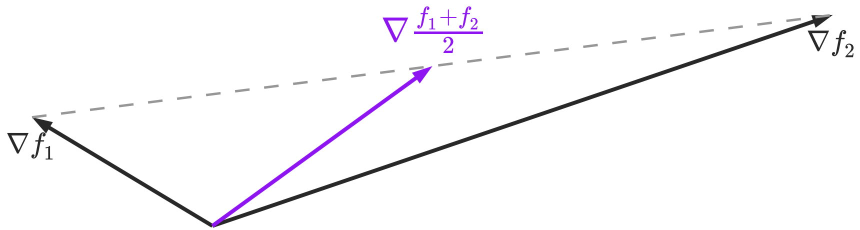
Motivation

Jacobian descent

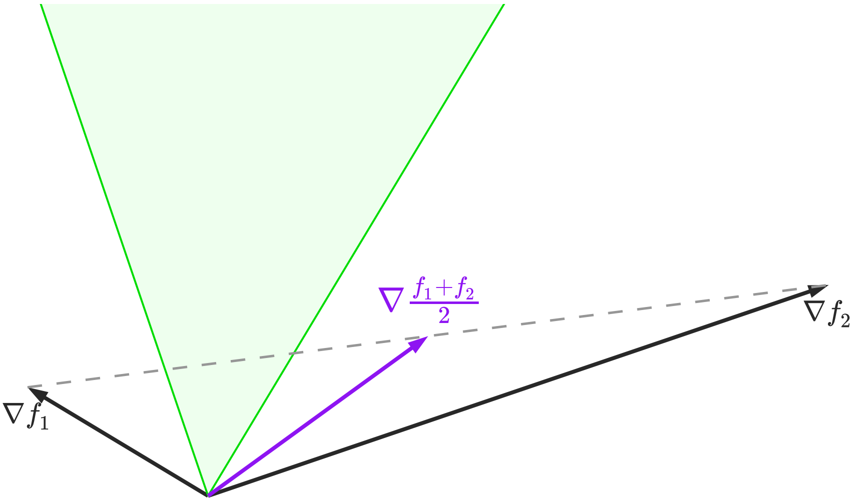
Jacobian descent



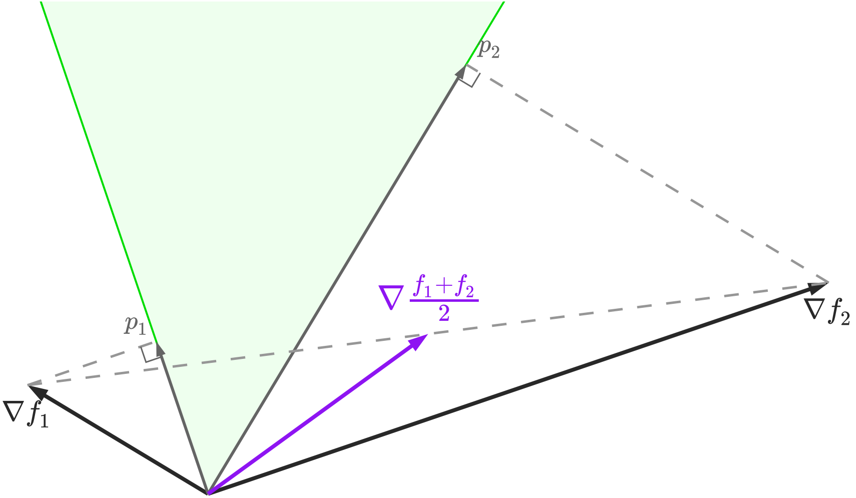
Jacobian descent



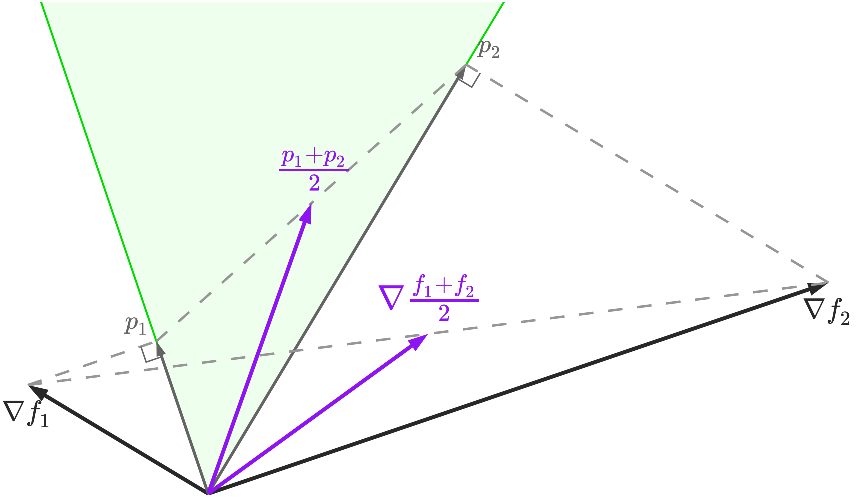
Jacobian descent



Jacobian descent



Jacobian descent



TorchJD

TorchJD: core functions

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`torchjd`	`autojac.jac_to_grad`	aggregate `.jac` fields into `.grad`

TorchJD: basic usage

```python
from torchjd import autojac
from torchjd.aggregation import UPGrad

batch_size = 16
loss_fn = MSELoss(reduction="none")  # do not average over the batch

for x, y in dataloader:
  z = model(x)  # shape: [16]
  losses = loss_fn(z, y)  # shape: [16]
  autojac.backward(losses)
  # parameters now have a .jac field
  autojac.jac_to_grad(model.parameters(), UPGrad())
  # parameters now have a .grad field
  optimizer.step()
  optimizer.zero_grad()
```

More examples at torchjd.org.

Discussion